

RESEARCH ARTICLE

Who Is the Best Helper for University Students' Mathematical Creativity? A Quasi-Experimental Study of Human–ChatGPT, Human–Google, and Human–Human Co-Creation

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ABSTRACT

Recent studies suggest that working with ChatGPT can generate more creative outcomes than humans alone. However, does ChatGPT retain its creative edge when humans have access to alternative information sources, such as Google Search or a human peer. This study addressed this question through a quasi-experiment with 230 Chinese university students in three groups (the human–ChatGPT group, the human–Google group, and human–human dyads) and a mathematical creativity task. The results showed that the human–ChatGPT group generated the most flexible and fluent solutions, while the human–human group produced the most original solutions in solving mathematical creativity tasks. The human–human group accurately assessed their actual performance, while the human–ChatGPT group overestimated it, and the human–Google group underestimated it. Furthermore, the study revealed that the human–Google group encountered greater difficulties, invested more effort, and reported lower levels of interest compared to the human–human and human–ChatGPT groups. Students found Google less useful than ChatGPT and their human peers. Similarly, students also found Google less effective than ChatGPT and their peers in enhancing self-efficacy. These findings highlight the benefits of human–human and human–ChatGPT co-creation in fostering mathematical creativity and call for further research on how to combine human inspiration with ChatGPT support to enhance it.

1 | Introduction

Mathematical open-ended tasks are widely used in education, particularly to foster students' creative thinking and problem-solving skills [1, 2]. These tasks often lack fixed solutions and

require students to engage in flexible reasoning and creative thinking within complex mathematical contexts [3–5]. With the advancement of educational theories and emerging technologies, mathematics education has increasingly emphasized enhancing students' creative thinking and problem-solving skills

through open-ended tasks [6]. As an emerging technology, ChatGPT can assist students in solving mathematical tasks by providing supporting information and generating prototypical solutions [7]. Recent studies have indeed found that human working with ChatGPT can generate more creative outcomes than humans alone. For example, Urban et al. [8] revealed that participants who used ChatGPT improved the quality, elaboration, and originality of their solutions. Habib et al. [9] also observed that students using ChatGPT scored higher in flexibility, fluency, and originality than those who did not use it. Similarly, Shaer et al. [10] showed that generative AI could enhance the idea generation and evaluation processes. However, a study comparing story writing found that AI-assisted stories were more like each other than those written by students alone [11]. Another study confirmed this, showing that human-generated solutions outperformed those generated by ChatGPT in terms of novelty, even when their quality was comparable [12]. Furthermore, the decline in novelty when co-creating with generative AI was observed in the arts [13]. Dell'Acqua et al. [14] also found that while participants using ChatGPT generated higher quality ideas, it reduced their diversity.

Based on these findings, there is reason to believe that human-ChatGPT co-creativity requires more rigorous investigation and should be compared with other co-creation processes, rather than being simply compared to individual human creativity. Since ChatGPT provides external cognitive support, research should also compare whether other sources of external knowledge can produce creative outputs like those of human-ChatGPT co-creation. It remains unclear whether ChatGPT retains its creative advantage when humans collaborate with other co-creation partners. This makes comparing creativity between humans and different co-creation partners an important research area. Although recent empirical studies have shown that human-human collaboration has greater potential than human-ChatGPT collaboration in creative generation and problem-solving (e.g., [15, 16]), to our knowledge, no research has yet compared different co-creation partners in solving mathematical creativity tasks (MCT). We posit that it is premature to conclude that human-ChatGPT co-creation always outperforms other forms of co-creation in solving creative tasks.

Therefore, this study aims to explore the impact of different co-creation partners on university students' performance (flexibility, fluency, and originality of solutions) in solving MCT through experimental design. For this purpose, this study sets up three experimental conditions, each working with a different co-creation partner: ChatGPT, Google Search, or a human peer. This will help us understand if non-ChatGPT external cognitive support enriches creativity in the same way as ChatGPT. Answering this question is crucial for theory and practice, as it will reveal whether human-ChatGPT co-creation is truly superior to other forms of co-creation. Furthermore, there are still aspects that need empirical exploration to better understand the boundary conditions and subtle differences among different forms of co-creation. This is precisely the gap this study also aims to address. Therefore, this study also examines five key factors essential for solving MCT: (1) accuracy of self-evaluation, (2) perceived task interest and difficulty, (3) mental effort invested, (4) perceived usefulness of partners, and

(5) self-efficacy for task resolution. Therefore, this study explores the following research questions:

RQ1: Do different co-creation partners influence creativity (flexibility, fluency, and originality) of participants' mathematical solutions?

RQ2: Do different co-creation partners influence participants' evaluation of their co-creation processes, such as the accuracy of self-evaluation, task interest and difficulty, invested mental effort, and their partner's usefulness?

RQ3: Do different co-creation partners affect participants' creative self-efficacy?

2 | Literature Review

2.1 | Mathematical Open-Ended Tasks

Solving mathematical open-ended tasks requires a combination of divergent and convergent thinking [4, 17]. Specifically, divergent thinking generates a variety of potential solutions or reasoning paths, providing a wealth of creative ideas [18]. Convergent thinking focuses on the decision-making process, helping students select the most reasonable and promising solutions from the many ideas generated [19]. These solutions are then evaluated to identify the final original solutions [20]. Therefore, a mathematical open-ended task is considered a MCT, as it combines both divergent and convergent thinking, encouraging individuals to develop original mathematical solutions [1].

2.2 | Co-Create to Solve Mathematical Open-Ended Tasks

Co-creating to solve MCT requires extensive interaction and coordination among collaborators [21]. While exchanging ideas, social (e.g., discussing viewpoints), motivational (e.g., competing), and cognitive (e.g., combining ideas) processes stimulate and enrich the co-creation process [22]. Moreover, since the co-creation process typically develops around a complex task, the task itself interests, difficulty, and invested mental effort also influence creative collaboration [23]. These task characteristics are often interrelated, for example, task difficulty and interest provide valuable cues to participants, guiding how they allocate and regulate cognitive resources in solving tasks (e.g., invested mental effort [24–27]). Also, task difficulty may be influenced by personal factors (e.g., self-efficacy, prior experience [28]). Although co-creation does not always result in more creative output, the activation of new knowledge and additional stimulation can outperform individuals who do not interact with others in creative idea generation [29], for example, through the variation and recombination of available knowledge, a core aspect of creativity [30, 31].

These mechanisms also apply to human-machine co-creation, particularly with ChatGPT, where the co-creation process is stimulated through collaboration [15]. ChatGPT is a generative

AI technology that produces content, such as texts or images, extrapolated from an extensive database [32, 33]. In this way, the co-creation process is enriched by the vast source of information and knowledge provided by ChatGPT. For example, participants can exchange ideas with ChatGPT in different ways [34] or use its knowledge to think more broadly and explore creative ideas interactively [35].

2.3 | Self-Efficacy in Co-Creation Processes

Creative self-efficacy refers to an individual's confidence in their ability to generate original ideas [36, 37]. If participants perceive a task as easier and more manageable with the support of their co-creation partners, they are likely to feel more creative, thus enhancing their creative self-efficacy [38]. As demonstrated in the research by Urban et al. [8], students using ChatGPT not only performed better in solving creative tasks but also found these tasks less challenging than their peers who worked alone. In the field of creativity, students with higher self-efficacy often tackle complex creative tasks and overcome challenges with greater confidence, as they perceive these tasks as manageable and within their capabilities [39, 40]. Therefore, they tend to produce better creative outcomes.

However, despite the improvement in idea generation from co-creating with a partner, participants may feel concerned that the partner is taking over the thinking process, potentially reducing their creative self-efficacy and engagement [9]. Similarly, McGuire et al. [41] showed that the level of influence participants have in the co-creation process with their partners affected their creative self-efficacy. In other words, participants with more control over the co-creation process reported higher levels of creative self-efficacy, while those with less control reported lower levels.

3 | Methods

3.1 | Procedure

This study was conducted in the middle of the academic semester. The corresponding author coordinated with the counselor to recruit students from all majors and announced the topic of the study ("Mathematical Creativity Tasks"). Students who volunteered for the study were rewarded with a few credits. Each student was randomly assigned to one of the three experimental conditions (human–ChatGPT group, human–Google group, or human–human group) using a lottery system. Before the experiment, students drew lots to determine which experimental condition they would participate in. This ensured that each student had an equal chance of being assigned to one of the conditions. After the draw, students were placed into the corresponding classrooms based on their random assignment.

Specifically, In human–ChatGPT group, students used personal computers with identical software and hardware settings. Although students were free to use the web browser for information searches, no one chose to do so. They only opened the ChatGPT through a single browser tab and were told that: (1) ChatGPT could respond to prompts; (2) providing more

background information could generate more accurate responses; (3) users could suggest modifications to further refine the response; (4) users could upload images or documents for processing. Similarly, participants in Human–Google group were only allowed access to Google Search, with all other AI tools and search engines being blocked. In human–human group, student pairs were randomly assigned and informed that one of them assumed a leadership role (person A). During task resolution, person A guided person B, who adopted a supportive and passive role without completing any work independently. Participants in three conditions were encouraged to draw inspiration from their partner (ChatGPT/Google Search/peer student will be referred to as partner in the following text). Each one worked in a separate and quiet area and could not see each other when working.

Before task resolution, informed consent and personal information forms were distributed to all participants. Subsequently, participants were given two experimental tasks. Finally, they received questionnaires designed to measure self-evaluation, task interest and difficulty, mental effort invested, their partner's contribution, and task self-efficacy.

All participants completed two experimental tasks (see Section 4 for details). First, all participants solved the baseline task alone to assess the mathematical creativity baseline. Second, three groups solved the MCT with the support of their partner. Each participant completed all the required tasks, and none were excluded. The entire study lasted 80 min ($SD = 7$) in human–ChatGPT group, 70 min ($SD = 5$) in human–Google group, and 68 min ($SD = 7.8$) in human–human group.

3.2 | Participants

A total of 230 Chinese university students participated in this study. All participants were homogenous in ethnicity and race (100% Chinese natives). In the present study, the human–ChatGPT group contained 60 participants, the human–Google group contained 60 participants, and the human–human group contained 110 participants (55 dyads). Since a priori sample size calculation was not performed, we conducted a post hoc power analysis using G*Power 3.1.9.7 software to assess the statistical power of this study based on the available sample size. The analysis assumed a medium effect size ($f = 0.25$), an alpha error probability ($\alpha = 0.05$), a correlation among repeated measures (Corr among rep measures = 0.5), and a nonsphericity correction ($\epsilon = 1$). The three statistical tests were performed: ANOVA: repeated measures, within-between interaction; ANOVA: repeated measures, between factors; and ANOVA: repeated measures, within factors. The results of the post hoc power analysis showed that the statistical power for the three tests was 0.99, 0.93, and 0.99, respectively, indicating that the sample size is sufficiently large to detect the expected effects.

As shown in Table 1, there were no significant differences among the three groups in age, gender, major, and prior ChatGPT experience. Additionally, all participants had no specific mathematics background and had achieved similar level of competence in prior mathematics courses. Therefore, all

TABLE 1 | Detailed information about participants.

	Human-ChatGPT (<i>N</i> = 60)	Human-Google (<i>N</i> = 60)	Human-Human (<i>N</i> = 110, 55 dyads)	Comparison
Age	<i>M</i> = 19.20, <i>SD</i> = 1.36	<i>M</i> = 19.27, <i>SD</i> = 1.10	<i>M</i> = 18.95, <i>SD</i> = 1.25	$F(2, 227) = 1.58$, $p = 0.208$
Gender				$\chi^2(2) = 4.43$, $p = 0.109$
Male	39	28	57	
Female	21	32	53	
Major				$\chi^2(4) = 6.76$, $p = 0.149$
Sciences	16	19	34	
Liberal arts	16	26	36	
Engineering	28	15	40	
Prior ChatGPT experience				$\chi^2(8) = 11.30$, $p = 0.185$
Never	2	2	1	
Seldom	10	7	18	
Sometimes	24	27	57	
Often	23	22	25	
Always	1	2	9	
Prior mathematical competence	<i>M</i> = 73.43, <i>SD</i> = 9.11	<i>M</i> = 75.05, <i>SD</i> = 9.46	<i>M</i> = 73.51, <i>SD</i> = 9.41	$F(2, 227) = 0.63$, $p = 0.536$

participants were homogeneous in mathematical competence, a vital predictor of mathematical creativity. In this study, mathematical competence was assessed with the paper-pencil mathematics test for personnel selection. This version was adapted from the official short form developed by Jasper and Wagener [42], which has been applied in Meier et al. [43].

The experiment followed all ethical guidelines applicable to human research participants and received ethics approval from the relevant ethical committee.

3.3 | Data Analysis

To address RQ1, that is, to assess the impact of different conditions on participants' flexibility, fluency, and originality, a mixed ANOVA was conducted. Specifically, flexibility, fluency, and originality were used as dependent variables, respectively, with condition as the between-subjects factor and time (pre, post) as the within-subjects factor. During the data analysis, Z-standardization was used to eliminate the influence of different test scales and items between pre-tests and post-tests. Subsequently, we performed post-hoc comparisons using Bonferroni correction to examine the group differences. Furthermore, analysis of covariance (ANCOVA) was employed with prior mathematical competence as a covariate. This allowed us to explore whether individual differences in prior mathematical competence influenced their mathematical creativity. Before conducting the ANCOVA, the data were screened for outliers, and assumptions including normality, homogeneity of variance, linearity, and homogeneity of regression slopes were tested and confirmed to be met.

To clearly address RQ2, the accuracy of self-evaluation across the different groups was first tested using ANOVA. Similarly, ANOVA analyses were conducted for task interest, difficulty, invested mental effort, and perceived partner helpfulness. To further evaluate the perceived contributions of participants and their partners, a mixed ANOVA was used, with condition as a between-subject factor and contributor (oneself, partners) as a within-subject factor, to analyze the interaction effects of perceived contributions across the different groups. Before conducting the ANOVA and mixed ANOVA, outlier screening was performed, and the assumptions of normality, homogeneity of variance, and independence were verified.

To address RQ3, a mixed ANOVA was further conducted, with self-efficacy as the dependent variable, condition as a between-subject factor, and time (pre, posttest) as a within-subject factor. This allowed us to examine the impact of different conditions on participants' self-efficacy over time. Before conducting the mixed ANOVA, outlier screening was performed, and the assumptions of normality, homogeneity of variance, and independence were verified.

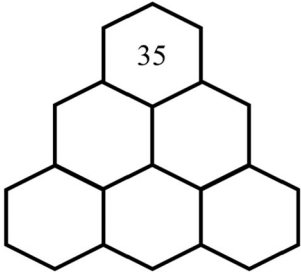
4 | Measures

4.1 | Mathematical Creativity Baseline

Before the experiment, the baseline task adapted from Kattou et al. [44] was used to measure the mathematical creativity baseline under three conditions. As shown in Table 2. Participants should generate as many original solutions as possible to fill the number pyramid.

TABLE 2 | Baseline task.

In the number pyramid, each regular hexagonal cell must contain a number, and each number can be computed by performing the same operation on the two numbers below it. When filling the pyramid, ensure that number 35 is at the top. Please find as many original solutions as possible.



4.2 | Mathematical Creativity Task

To measure mathematical creativity performance, the MCT was adapted from Lee and Seo [45]. The original MCT comprised five mathematical open-ended tests designed to assess three key indicators of mathematical creativity: flexibility, fluency, and originality. The MCT has been widely adopted in previous studies as a valid measure of mathematical creativity [46–48]. Notably, one test (regular hexagon problem) was excluded from the adapted version due to its low difficulty index [49]. The adapted MCT underwent internal consistency testing, yielding results that demonstrated very high reliability. This version includes the following four tasks (as shown in Figure 1):

Task 1: Draw as many 2 cm² figures as possible within a 4 × 4 grid (where the distance between adjacent points is 1 cm). If a figure overlaps after rotation or inversion, it is considered the same and should only be counted once. All figures should be complete and independent, without sharing any points or line segments.

Task 2: Devise as many methods as possible to measure the degree of dispersion of five marbles thrown by three students.

Task 3: When a rectangular prism containing water is tilted, the geometric shape formed at the interface between the water surface, and the flask surface also changes. Students should identify as many properties as possible related to the size and shape of the water.

Task 4: It includes eight geometries, and the goal is to identify as many shapes as possible that share the same features as shape A, and to describe these common characteristics.

4.3 | Self-Evaluation

According to the study by Karwowski et al. [50], all participants evaluated the creativity (e.g., flexibility, fluency, originality) of their mathematical solutions after task resolution. Specifically, the statements for flexibility were as follows: Please rate how many different correct solutions you considered in solving the task on a scale of 1–100. The statements for fluency were as follows: Please rate the number of correct answers in each solution on a scale of 1–100. The statements for originality were as follows: Please rate the rarity of your solution on a scale of 1–100.

4.4 | Accuracy of Self-Evaluation

Accuracy of self-evaluation (bias index) was used to measure the gap between self-evaluation and actual performance [51]. It was calculated by dividing the difference between the self-evaluation score (i.e., flexibility, fluency, and originality) and the actual creativity scores (converted to a 1–100 scale) by 100. Specifically, actual creativity scores were converted to a 1–100 scale based on percentiles within each experimental group (human–ChatGPT group, human–Google group, or human–human group), rather than using percentiles from the whole sample. This approach ensures that each student’s accuracy of self-evaluation is measured relative to their performance within their own experimental group, avoiding potential cross-group differences that could affect the results. The bias index ranges from –1 to 1, where negative values represent underestimation of creativity, positive values represent overestimation of creativity, and values close to zero represent accurate self-evaluation.

4.5 | Perceived Interest

After completing the task, all participants assessed task interest by answering the following statement [52]: Please rate your interest in solving this task on a scale of 1 (very boring) to 100 (very interesting).

4.6 | Perceived Difficulty

After completing the task, all participants assessed task difficulty by answering the following statement [53]: Please rate the difficulty in solving this task on a scale from 1 (very easy) to 100 (very difficult).

4.7 | Perceived Mental Effort

After task resolution, all participants assessed invested mental effort by answering the following statement [54]: Please rate the mental effort invested in solving this task on a scale from 1 (no effort) to 100 (full effort).

4.8 | Perceived Partner Usefulness

Participants in the three groups rated the usefulness of partner by answering the following statement [8]: Please rate

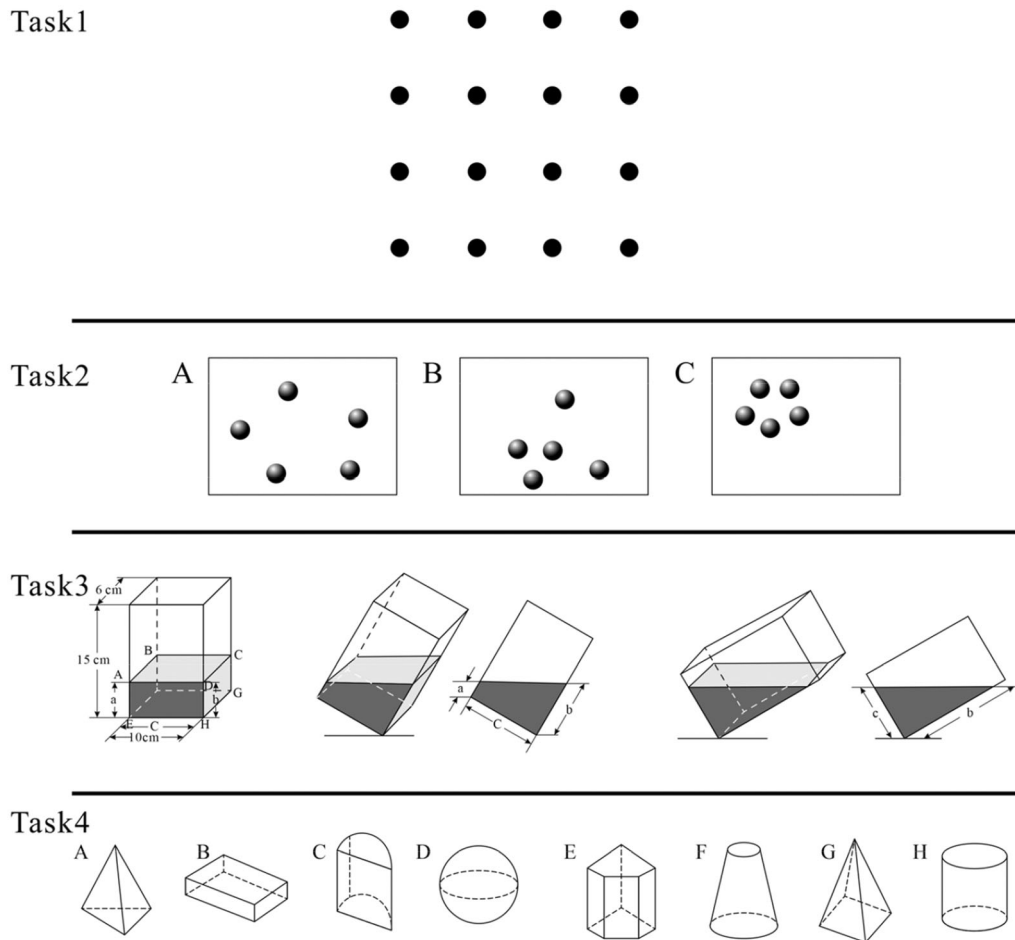


FIGURE 1 | Images of MCT. These images were shown to students in the task. *Source:* Images of MCT were adapted from García-García et al. [46].

the usefulness of ChatGPT/Google Search/peer student in solving this task on a scale from 1 (very useless) to 100 (very useful).

Moreover, participants in the three groups rated the contribution of oneself and their partner by answering the following statement [15]: Please rate the contribution of yourself and ChatGPT/Google Search/peer student in solving this task on a scale from 1 (no contribution at all) to 100 (very high contribution). Participants' ratings were not required to add up to 100%.

4.9 | Self-Efficacy

After reading the MCT for the first time but before completing the task, all participants rated three statements (e.g., I am confident in generating creative solutions to construct as many 2 cm^2 figures as possible [55]) on a scale ranging from 1 (completely disagree) to 100 (completely agree). After task resolution, participants in the three groups also rated three statements (e.g., I am confident that I can measure the extent of marble scatter with the help of ChatGPT/Google Search/peer student). The reliability of the statements was strong, with Cronbach's α ranging from 0.90 to 0.96 in the pretest and from 0.88 to 0.95 in the posttest.

5 | Mathematical Creativity Scoring

The assessment of mathematical creativity was based on the flexibility, fluency, and originality of students' solutions [44, 45]. Flexibility refers to the number of correct answer categories in each task. Fluency refers to the number of correct answers in each category, with a maximum of five points in each category. Originality refers to the rarity of the answer. We calculated the proportion of each answer category relative to the total number of responses. If the proportion of an answer category falls within (0, 1%), (1%, 2%), or (2%, 3%), the scores were three, two, and one, respectively. Therefore, we first marked the students' answers in the baseline task and the MCT (four tasks) as correct (one) or wrong (zero). Then, we categorized the correct answers. Finally, based on the above scoring criteria, we assigned scores for the flexibility, fluency, and originality of answers in each task. The scores of three indicators were summed to measure each student's baseline score in mathematical creativity. For the MCT, the score for each task was calculated by evaluating flexibility, fluency, and originality of solutions. The total score for a student's three indicators was obtained by summing the scores from the four tasks. The scoring reference tables (see Supporting Information) were adapted from Lee and Seo [45].

Before scoring the entire task, the research team independently analyzed and scored 18 pre-study samples, achieving an initial

interrater reliability with a Cohen's kappa of 0.85, indicating substantial agreement. The research team then reviewed the discrepancies to reinforce the consistency of the scoring standards. After the scoring exercise, the two scorers independently scored the remaining data, with a Cohen's kappa exceeding 0.91, reflecting almost perfect agreement. The research team reanalyzed all discrepancies until a 100% agreement was reached.

6 | Results

The results section examines whether different co-creation partners lead to different levels of flexibility, fluency, and originality of participants' mathematical solutions; whether different co-creation partners affect the accuracy of self-evaluations, perceived task interesting and difficulty, invested mental effort, and perceived partner usefulness. Furthermore, it also investigates whether different co-creation partners result in varying levels of participants' self-efficacy. Table 3 presents the descriptive statistics for the three groups, along with the ANOVA results.

6.1 | Co-Creation With Different Partners Affects the Flexibility, Fluency, and Originality of Solutions (RQ1)

The results of the mixed ANOVA showed that the main effect of time was not significant for flexibility, $F(1, 172) = 0.01$,

$\eta^2p = 0.00$, $p = 0.907$, but the main effect of condition was significant, $F(2, 172) = 3.80$, $\eta^2p = 0.04$, $p < 0.05$, and there was a moderate interaction effect, $F(2, 172) = 6.08$, $\eta^2p = 0.07$, $p < 0.01$. Further Bonferroni post-hoc comparisons showed that the flexibility of the human-ChatGPT group increased moderately, $t(59) = 2.48$, $p < 0.050$, $d = 0.44$. The flexibility of the human-human group also increased moderately, $t(54) = 2.24$, $p < 0.050$, $d = 0.36$, while the human-Google group did not change significantly, $t(59) = 0.71$, $p = 0.483$, $d = 0.11$.

For fluency, the main effect of time was not significant, $F(1, 172) = 0.01$, $\eta^2p = 0.00$, $p = 0.921$, but the main effect of condition was significant, $F(2, 172) = 3.31$, $\eta^2p = 0.04$, $p < 0.05$, and there was a small interaction effect, $F(2, 172) = 4.68$, $\eta^2p = 0.05$, $p < 0.05$. Bonferroni post-hoc tests revealed that the fluency of the human-ChatGPT group increased moderately, $t(59) = 2.14$, $p < 0.050$, $d = 0.39$, while the human-Google group did not change significantly, $t(59) = 0.80$, $p = 0.426$, $d = 0.13$. Also, the fluency of the human-human group did not change significantly, $t(54) = 1.90$, $p = 0.064$, $d = 0.31$.

For originality, as shown in Figure 2, there was no difference in mathematical creativity among the three groups in the baseline task, $F(2, 172) = 0.21$, $\eta^2p = 0.00$, $p = 0.808$. Furthermore, the results of the mixed ANOVA showed that the main effect of time was not significant, $F(1, 172) = 0.02$, $\eta^2p = 0.00$, $p = 0.883$, but the main effect of condition was significant, $F(2, 172) = 8.56$, $\eta^2p = 0.09$, $p < 0.001$, and there was a moderate interaction effect, $F(2, 172) = 8.19$, $\eta^2p = 0.09$, $p < 0.001$. Further Bonferroni post-hoc comparisons showed that the

TABLE 3 | Descriptive statistics and ANOVA results for individual variables among the three groups.

	Human-ChatGPT		Human-Google		Human-human		$F(2, 172)$	η^2p	p
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>			
1. Creativity baseline	15.53	5.70	15.15	6.48	15.91	6.51	0.21	0.00	0.808
2. MCT (flexibility)	15.03	3.78	12.83	3.29	12.38	3.35	9.71	0.10	< 0.001
3. MCT (fluency)	27.85	7.56	23.78	6.50	23.42	5.93	7.91	0.08	0.001
4. MCT (originality)	5.47	2.83	3.88	2.32	7.00	3.10	18.31	0.18	< 0.001
5. Self-evaluation (flexibility)	71.33	17.19	61.47	20.60	61.45	20.16	5.13	0.06	0.007
6. Self-evaluation (fluency)	73.05	14.51	61.00	13.12	63.87	16.65	10.84	0.11	0.000
7. Self-evaluation (originality)	61.02	16.42	34.13	17.20	53.76	16.73	41.03	0.32	0.000
8. Bias index (flexibility)	0.09	0.23	0.00	0.18	0.00	0.20	3.59	0.04	0.030
9. Bias index (fluency)	0.08	0.23	-0.05	0.18	-0.01	0.20	6.71	0.07	0.002
10. Bias index (originality)	0.15	0.29	-0.05	0.26	0.00	0.28	8.67	0.09	< 0.001
11. Interest	77.80	10.64	68.00	13.23	79.76	11.46	16.62	0.16	< 0.001
12. Difficulty	62.67	12.64	75.63	10.57	61.76	11.38	26.55	0.25	< 0.001
13. Effort	78.42	10.48	84.73	10.87	76.78	12.13	8.29	0.09	< 0.001
14. Usefulness of partner	74.77	13.90	62.08	11.71	72.09	12.70	16.27	0.16	< 0.001
15. Own contribution	32.15	11.24	69.13	9.21	48.22	8.66	214.87	0.71	< 0.001
16. Partner's contribution	67.15	9.25	30.82	9.80	48.71	9.51	218.45	0.72	< 0.001
17. Self-efficacy (pre)	47.81	20.76	42.77	22.27	46.04	20.00	0.88	0.01	0.416
18. Self-efficacy (post)	56.88	20.92	44.88	21.52	68.82	19.58	19.17	0.18	< 0.001

Note: *p*-values in this table have been corrected using Bonferroni correction to control Type I errors due to multiple comparisons. Abbreviation: MCT, mathematical creativity tasks.

originality of the human–ChatGPT group did not change significantly, $t(59) = 0.12$, $p = 0.906$, $d = 0.02$, the human–Google group decreased moderately, $t(59) = 3.14$, $p < 0.010$, $d = 0.48$, and the human–human group increased moderately, $t(54) = 2.73$, $p < 0.010$, $d = 0.45$. These findings indicated both between-subject differences (the human–human group achieved greater originality than the other groups) and within-subject differences (individual participants generated more original solutions when working with a peer than without). Figure 3 presents two examples from each group in Task 1.

Surprisingly, the results of the ANCOVA showed that prior mathematical competence was unrelated to flexibility, $F(2, 171) = 0.02$, $\eta^2 p = 0.00$, $p = 0.880$, fluency, $F(2, 171) = 0.05$, $\eta^2 p = 0.00$, $p = 0.822$, and originality, $F(2, 171) = 0.40$, $\eta^2 p = 0.00$, $p = 0.529$, in solving MCT.

6.2 | The Accuracy of Self-Evaluation Is Affected by Different Co-Creation Partners (RQ2)

The accuracy of self-evaluation showed a small and significant difference among the three groups in flexibility, $F(2, 172) = 3.59$, $\eta^2 p = 0.04$, $p < 0.05$, and a moderate difference in fluency, $F(2, 172) = 6.71$, $\eta^2 p = 0.07$, $p < 0.01$, and originality, $F(2, 172) = 8.67$, $\eta^2 p = 0.09$, $p < 0.001$. Specifically, in terms of flexibility, fluency, and originality, the bias index was close to zero for the human–human group, positive for the human–ChatGPT group, and negative for the human–Google group. In other words, the human–human group accurately assessed their actual performance (flexibility, fluency, and originality of the solutions are hereafter labeled as actual performance), while the human–ChatGPT group tended to overestimate it, and the human–Google group underestimated it.

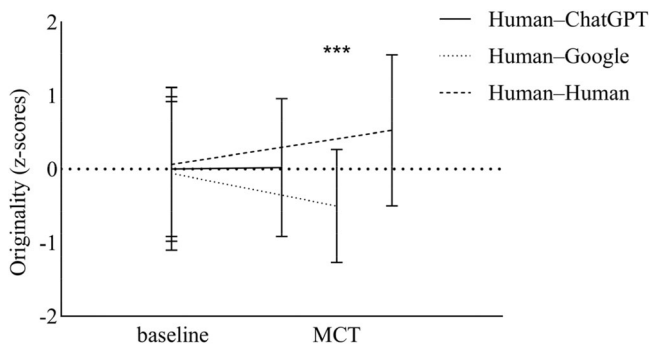


FIGURE 2 | Originality of solutions in baseline task and MCT. MCT: Mathematical creativity tasks. *** $p < 0.001$

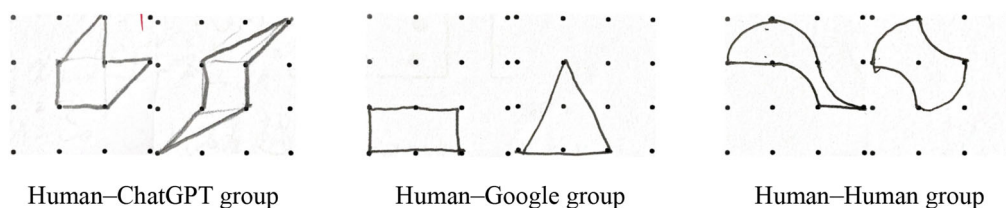


FIGURE 3 | Two examples for each group in Task 1.

6.3 | Task Interesting, Difficulty, Invested Mental Effort, and Partner's Contribution Are Significantly Affected by Different Co-Creation Partners (RQ2)

There was a strong and significant difference in task interest, $F(2, 172) = 16.62$, $\eta^2 p = 0.16$, $p < 0.001$, and difficulty, $F(2, 172) = 26.55$, $\eta^2 p = 0.25$, $p < 0.001$, as well as a moderate and significant difference in mental effort invested, $F(2, 172) = 8.29$, $\eta^2 p = 0.09$, $p < 0.001$, among the three groups. As further illustrated in Figure 4. Compared to the human–ChatGPT and human–human groups, the human–Google group encountered more difficulties, invested more effort, and reported lower levels of interest.

Furthermore, participants in the three conditions showed a strong and significant difference in perceived partner helpfulness, $F(2, 172) = 16.27$, $\eta^2 p = 0.16$, $p < 0.001$. As shown in Figure 5, the human–ChatGPT group and human–human group perceived their partner as similarly helpful, while participants in human–Google group perceived Google Search as useless. These findings were further explored through the evaluations of participants' and their partners' contributions to the task. The results of the mixed ANOVA revealed a strong and significant interaction effect, $F(2, 172) = 407.27$, $\eta^2 p = 0.83$, $p < 0.001$. In the human–human group, the perceived contribution of both sides was equal, each close to 50%. However, participants in the human–Google group assessed their own contributions as strongly higher than their partners, whereas participants in the human–ChatGPT group attributed most of the contribution to their partners.

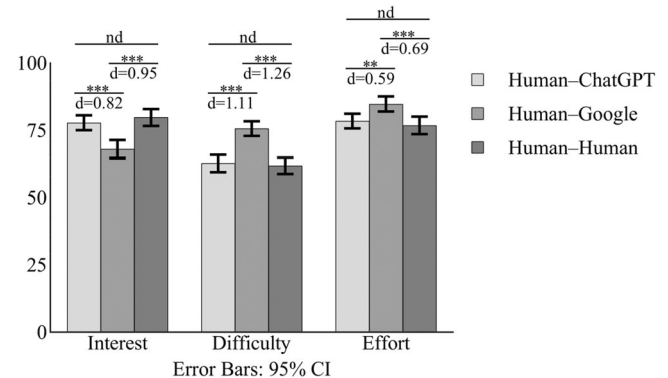


FIGURE 4 | Perceived interest, difficulty, and invested mental effort. Source: Holm's correction was used for the post hoc pairwise comparisons. d = Cohen's d ; nd = nonsignificant differences. *** $p < 0.001$, ** $p < 0.010$.

6.4 | Self-Efficacy for Task Resolution Is Influenced by Different Co-Creation Partners (RQ3)

The results revealed strong and significant main effects for condition, $F(2, 172) = 11.94$, $\eta^2 p = 0.12$, $p < 0.001$, and time, $F(1, 172) = 27.03$, $\eta^2 p = 0.14$, $p < 0.001$, as well as a moderate interaction effect, $F(2, 172) = 7.60$, $\eta^2 p = 0.08$, $p < 0.01$. Post hoc tests with Holm's correction revealed a strong and significant increase in self-efficacy from pretest to posttest in the human-human group, $t(54) = 6.04$, $p < 0.001$, $d = 1.15$, and a small improvement in the human-ChatGPT group, $t(59) = 2.48$, $p < 0.050$, $d = 0.44$. Participants' self-efficacy did not change significantly in the human-Google group, $t(59) = 0.55$, $p = 0.588$.

Thus, as shown in Figure 6, while there was no difference in self-efficacy in the pretest, $F(2, 172) = 0.88$, $\eta^2 p = 0.01$, $p = 0.416$, posttest assessments showed a strong difference, $F(2, 172) = 19.17$, $\eta^2 p = 0.18$, $p < 0.001$. Specifically, participants in the human-human group exhibited moderately higher self-efficacy than those in the human-ChatGPT group ($d = 0.59$), and participants in the human-ChatGPT group

showed strongly higher self-efficacy than those in the human-Google group ($d = 0.57$).

7 | Discussion

The present study explored the processes and outcomes of collaborative interactions between humans and ChatGPT. Unlike previous studies that primarily compared creative ideas generated separately by humans and ChatGPT (e.g., [56–58]), this study focused on comparing different forms of co-creation: human-ChatGPT, human-Google, and human-human. Our reasoning was that relying on a simple comparison of ideas generated by humans and ChatGPT might be inconclusive, given the vast databases large language models are trained on and have access to ChatGPT, which are obviously unavailable to individual participants. Thus, we introduced an additional control group in our study, where participants co-creation with Google, allowing them to identify potential solutions that would be unavailable when working alone. Given that the human-ChatGPT and human-Google groups involved cooperation between humans and external enhancement tools, an appropriate control condition was not individual participants, but human-human dyads, allowing participants to exchange perspectives and share information. This study explored multiple aspects of the co-creation process. Therefore, we not only explored the creativity of participants working with different co-creation partners, but also analyzed their accuracy in self-evaluation, perceived task interest and difficulty, invested mental effort, creative self-efficacy, and perceived partner usefulness. Next, we will discuss the main findings of the current investigation.

7.1 | Human-ChatGPT Co-Creation Improves Flexibility, Fluency of Solutions, While Human-Human Enhances Originality

This study found that participants who used ChatGPT generated solutions with greater flexibility and fluency compared to those generated through collaboration with a peer or Google. However, participants working with peers produced the most original solutions among the three groups. This suggests that ChatGPT can enhance the diversity and quantity of solutions in solving MCT, but it does not increase their rarity. These findings align with existing studies, which suggest that ChatGPT may enhance productivity but not necessarily idea originality [11, 12, 14]. This phenomenon can be explained through the perspective of computational creativity, where AI tools like ChatGPT use vast amounts of data to propose various possibilities based on precedents in the knowledge base [59]. This ability to quickly iterate existing ideas and offer diverse solutions may foster an environment that is conducive to creative thinking, enhancing flexibility and fluency. Also, the rapid idea generation, without the limitations of human fatigue or cognitive biases, enables participants to produce more responses in a shorter time, leading to higher scores compared to human-human or human-Google groups [60]. In other words, the flexibility and fluency observed in the human-ChatGPT group may be attributed to its ability to integrate various elements

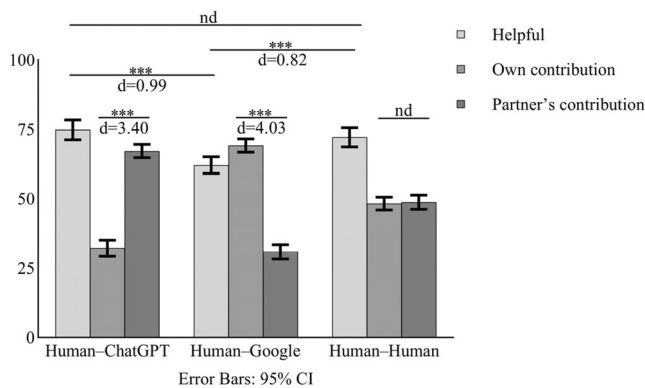


FIGURE 5 | Perceived usefulness of partners. Source: Holm's correction was used for the post hoc pairwise comparisons. d = Cohen's d ; nd = nonsignificant differences. *** $p < 0.001$.

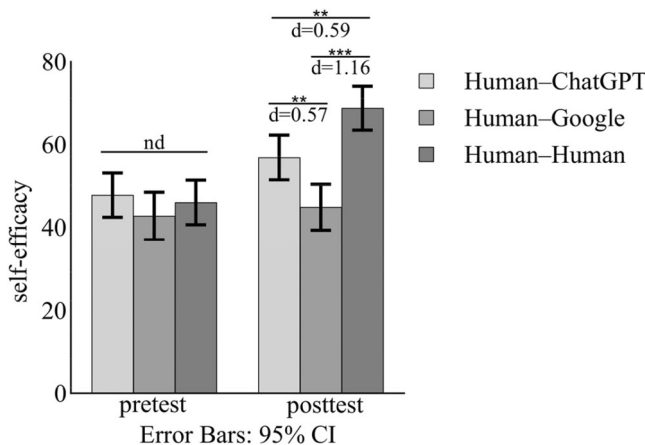


FIGURE 6 | Self-efficacy in the pretest and posttest. Source: Holm's correction was used for the post hoc pairwise comparisons. d = Cohen's d ; nd = nonsignificant differences. *** $p < 0.001$, ** $p < 0.010$.

from the existing mathematical solution pool and rapidly generate multiple potential solutions, thereby broadening participants' cognitive perspective and encouraging them to explore diverse approaches. Therefore, ChatGPT is most effective at generating incrementally new ideas, rather than radically new ones [61].

Furthermore, this finding highlights the crucial role of the social and interactive nature of human–human co-creation in fostering creative thinking, as human–human collaboration is more likely to involve reflective and critical discussions that challenge conventional thinking [15]. This is further supported by Song et al. [62], who found that when discussing a topic, student peers may naturally initiate new lines of inquiry or ask questions, due to their human ability to interpret social cues and their possession of personal interests. This may encourage them to engage in a dynamic exchange of ideas [16, 63]. These abilities are advantages of human interactions that are beyond the reach of current chatbot technology. In other words, human–human co-creation challenges each other's thinking and builds upon one another's insights, leading to the generation of novel and unique solutions. However, working with a peer may bring its own set of limitations, such as divergent perspectives, communication barriers, or the inherent cognitive biases of both individuals, which could limit the quantity of potential solutions [64]. While peer collaboration allows for a more dynamic exchange of ideas, it may also lead to group thinking or a focus on solutions that align with individual's prior knowledge or experiences, which could restrict fluency.

Moreover, this study highlighted the limitations of using Google as a collaborative tool. While Google provides access to a vast amount of information, it may not offer the same level of personalized feedback and interactive support as ChatGPT or human peers [65]. Similarly, Google cannot effectively integrate information like ChatGPT, thereby facilitating creative thinking at the same level. Its reliance on keyword matching and existing content may limit the generation of original solutions, especially when students are not adept at using the tool for solving MCT. These suggest that simply obtaining information, without the interactive and reflective elements of collaboration, may not effectively enhance creativity. This finding aligns with some studies [66–68], suggesting that simply acquiring fragmented information from Google does not necessarily translate into learning outcomes; instead, the way information is processed and discussed is important.

Overall, while ChatGPT enhances flexibility and fluency of solutions in MCT, human–human interactions are essential for fostering originality. Therefore, a teaching approach that balances the advantages of human–ChatGPT and human–human co-creation may be the most effective way to enhance creative thinking in educational settings [69]. Educators should consider integrating ChatGPT into the classroom to support creative thinking and offer diverse perspectives. Teachers should also provide opportunities for human–human collaboration to foster originality and promote deeper cognitive engagement. Furthermore, since our experimental conditions involved dyads: either participants working with machines or human peers, our findings are not directly comparable to studies with individuals (e.g., [8]). Future research should directly compare different

dyadic collaboration formats to better understand how interaction dynamics (whether human–human or humans–ChatGPT co-creation) affect creative performance. Such comparisons would help identify whether the observed effects are primarily due to the nature of the collaborator (ChatGPT vs. human) or the collaborative format itself.

7.2 | Prior Mathematical Competence Was Not Related to Mathematical Creativity

The findings indicated no significant relationship between mathematical competence and mathematical creativity, aligning with Meier et al.'s (2024) research, but contrasting with others that reported a positive correlation (e.g., [70]). One possible explanation is that mathematical creativity relies not only on the acquisition of prior mathematical knowledge but also on the ability to overcome mental sets during problem-solving [71]. For instance, tasks such as generating multiple solutions or posing new problems require divergent thinking, which may be more strongly influenced by noncognitive factors (e.g., creative self-beliefs, personality traits) than by domain-specific knowledge alone [72]. Moreover, mathematical competence is often assessed through standardized tests or academic performance, which tend to emphasize procedural fluency and factual knowledge rather than creative thinking. Thus, individuals who perform well on traditional math assessments may not necessarily excel in tasks that demand innovation [73].

Interestingly, Meier et al. [74] also found that although there was some association between mathematical competence and mathematical creativity, the relationship was modest and inconsistent across samples and measures. This suggests that mathematical creativity may represent a relatively distinct construct, with developmental mechanisms that differ from those underlying general mathematical achievement. Overall, the current findings reinforce the idea that mathematical creativity is not solely determined by prior mathematical competence. Future work should further explore the role of noncognitive variables—such as creative self-efficacy, mindset, and affective factors like math anxiety—in shaping mathematical creativity.

7.3 | The Human–Human Group Accurately Assesses Creativity, While the Human–ChatGPT Group Overestimates and the Human–Google Group Underestimates It

Students' self-evaluation plays a crucial role in the problem-solving process [75]. When students assess their own work, they are asked to reflect on the strategies they used and the overall quality of their work [76]. This study found that the human–human group accurately assessed their actual performance, while the human–ChatGPT group overestimated it, and the human–Google group underestimated it. Furthermore, significant differences were observed in the absolute accuracy of self-evaluation among the three groups. This finding can be inspired by previous research, for example,

Zamfirescu-Pereira et al. [77] found that novice users were inaccurate in their self-evaluation when working with a chatbot. Molenaar [78] also found that young scholars tended to overestimate their performance when using adaptive learning technologies, especially when their performance was not immediately evident or when feedback was delayed. This may be linked to the lack of accurate feedback mechanisms and ineffective self-regulation. Similarly, Urban et al. [8] revealed that the relationship between individual performance and self-evaluation is not simply linear; rather, it is influenced by the self-evaluation group to which the participants are assigned. As Dörrenbächer-Ulrich and Perels [79] have shown that participants can be broken down into those who are skilled but underestimate their performance, those who are unskilled and overestimate their performance, those who are skilled and aware, and those who are unskilled but aware. In other words, participants' self-evaluations may vary across groups, as individuals differ in their skill levels and awareness of their self-evaluation.

Furthermore, from an exploratory perspective, it is worth noting that higher self-evaluation was strongly correlated with the perceived usefulness of ChatGPT (see Appendix: median $r = 0.34$, $r_{\text{bias index}} = 0.38$). This may be important because the perceived usefulness of ChatGPT was not related to actual mathematical creative (median $r = -0.04$). This suggests that participants who overestimate the usefulness of ChatGPT may become overly reliant on it, which could lead them to overestimate their own performance.

7.4 | Participants' Perception of Collaborating With Different Partners in the Co-Creation Process

This study found that students working with Google experienced greater difficulty, invested more effort and reported lower levels of interest than those working with ChatGPT or a peer. This perception may stem from the relatively poor user experience provided by traditional search engines like Google Search when compared to large language models such as ChatGPT, which has sparked discussions about the inevitable evolution of traditional search engines into thinking machines through the integration of search functionality and large language models [80, 81]. Furthermore, students working with ChatGPT or peers saw their partner as extremely helpful. When students were asked about their own and their partner's contributions, a much-telling interaction was observed. Specifically, the perceived contribution of both individuals in dyads was nearly equal (around 50%). Participants in the human-Google group perceived their role as much more dominant than their partner, while in the human-ChatGPT group, the opposite effect was observed with a large effect size: here ChatGPT was clearly leading. Although based on declarative data, the observed patterns support the internal validity of our investigation and provide potential explanations for participants' self-perceptions (perceived task interest, difficulty, and invested mental effort) of the co-creation process. If students perceive ChatGPT as the main contributor to the creative process (believing ChatGPT is useful), how might this influence their actual performance? We explore the question in the remaining part.

Participants' self-evaluation judgments rely on different heuristic cues rooted in task characteristics and their self-perceptions [82, 83]. The question is that there was no significant correlation between the self-evaluation of participants using ChatGPT and their actual performance (see Appendix Tables A1). The explorations in this study provide two cues. First, the perceived usefulness of ChatGPT was strongly correlated with higher self-evaluation (see Appendix Tables A1, $r_{\text{bias index}} = 0.38$). This means that the more useful participants perceived ChatGPT to be, the more they overestimated their actual performance. Second, perceived task difficulty was positively associated with invested mental effort (see Appendix Tables A1, $r = 0.54$). In other words, participants who found the task easier invested less mental effort. Based on these exploratory findings, it can be inferred that when participants perceived ChatGPT as more useful, they were likely to overestimate their mathematical creativity. This overestimation may increase their satisfaction with the current solutions, which in turn encourages them to invest more effort—even when they perceive the task as somewhat challenging. These findings suggest that students who used ChatGPT believe they are performing well and investing greater effort, yet this may not translate into actual performance, as the effectiveness of subjective effort appraisals can be influenced by cognitive biases or other factors [26]. Therefore, future work should explore ways to enhance mathematical creativity while preventing overestimation. Furthermore, researchers should focus on designing more effective tasks that allow participants to accurately assess their creativity, thereby minimizing the potential impact of inaccurate self-evaluation.

7.5 | Human-ChatGPT and Human-Human Co-Creation Enhances Task Self-Efficacy

In the present study, participants' creative self-efficacy was measured twice: once before collaborative creativity and once after task resolution. Although the initial levels of creative self-efficacy were comparable across conditions, participants in the human-human group showed a significant increase in creative self-efficacy. Participants in the human-ChatGPT group exhibited a slight improvement, whereas those in the human-Google group showed no change. The slight improvement in self-efficacy observed in the human-ChatGPT group may be explained by research related to task-specific creative self-efficacy. For example, Urban et al. [8] showed that while participants using ChatGPT reported higher self-efficacy, the accuracy of their self-perception was affected by the perceived usefulness of ChatGPT. This finding raises concerns about the reliability of self-evaluation in ChatGPT-assisted learning, as perceived usefulness may be misinterpreted as evidence of ability. Thus, the higher self-efficacy in the human-ChatGPT group may reflect emotional responses or heuristic judgments rather than an accurate evaluation of creativity. However, since Urban et al. [8] did not focus on human-human collaboration, comparisons with our findings should be made cautiously. Although preliminary and requiring replication and further investigation into the exact mechanisms, the increase in creative self-efficacy through dyadic collaboration, where individuals can exchange ideas and share knowledge, may be theoretically and practically sound. As confirmed by the

findings of Tang et al. [15], dyadic collaboration creates a supportive and motivating environment that boosts individuals' confidence in their creative abilities. Creative self-efficacy is a subset of creative agency, so observing its growth within dyadic interaction is promising [39]. Therefore, this study opens new avenues for theorizing how creative agencies develop within dyads, groups, and teams.

Furthermore, from an exploratory perspective, it is worth noting that actual performance and self-efficacy (see variables three, four, and five in Appendix Tables A1 and A3) showed an ignorable correlation in the human–ChatGPT group (median: $r_{\text{pre}} = 0.07$, $r_{\text{post}} = 0.04$). However, participants in the human–human group showed a moderate positive correlation between self-efficacy and the originality of their solutions ($r_{\text{pre}} = 0.29$, $r_{\text{post}} = 0.29$). These findings suggest that higher self-efficacy may not benefit mathematical creativity in the human–ChatGPT group, but it may enhance originality in the human–human group. This can be explained by Schunk and Mullen's [84] finding, which indicates that when aggregated for all participants, higher self-efficacy is associated with better performance. Specifically, participants with higher self-efficacy are more likely to focus their attention on achieving their goals and seeking solutions from various perspectives. Conversely, those with lower self-efficacy lack the confidence to develop creative solutions, assuming that the answers are already known. However, at the within-subject level, higher self-efficacy may be negatively correlated with performance [55], as participants using ChatGPT tend to believe they will perform better on certain tasks than on others. This belief may hinder their performance on tasks they believe they will excel at.

8 | Limitations and Future Research

This study has several limitations, primarily due to its experimental nature. The study employed a MCT that was influenced by factors such as participants' mathematical beliefs, motivations, and attitudes, which could potentially impact their creativity on the test. Although known differences between the groups (e.g., creativity baseline) were controlled, the study did not account for all potential confounding variables, especially those factors that were not considered in experimental design. Therefore, while the intervention effects of different co-creations were observed, unmeasured psychological factors may still serve as potential drivers of the observed differences. Future research should strengthen experimental designs by incorporating more comprehensive baseline measurements, using advanced matching techniques to create more comparable groups, and employing multiple control groups to identify potential confounders.

Second, the frequency of prior ChatGPT usage was used as a measure of participants' familiarity with the tool in this study. However, this is a simplistic measure that may not fully capture the diversity of ways participants engage with generative models, especially in tasks involving mathematical creativity. Future work could benefit from assessing how participants use ChatGPT in specific contexts, such as exploration, idea generation, or refinement. Also, a more detailed approach could include behavioral data or qualitative reflections on how

participants interact with the tool. This would provide a deeper understanding of ChatGPT's role in fostering mathematical creativity and mathematical problem-solving.

Moreover, it should be acknowledged that the adapted MCT used in this study focused more on geometry than on other areas of mathematics. Future research should develop a MCT that covers a broader range of content areas and evaluates students' creativity across different mathematical domains. Future work should also include a more diverse and representative sample to enhance the external validity of the results, as the current sample was drawn from a limited pool. This expansion (e.g., different educational levels and backgrounds) can offer important insights into the potential differences in the effects of different co-creations, enhancing the generalizability of the research.

Finally, excessive reliance on ChatGPT led participants to adopt fixed problem-solving strategies, which may hinder students' intrinsic creativity. As observed, participants using ChatGPT spend more time resolving tasks compared to the other two groups. Future research should focus on the long-term impact of using ChatGPT on students' creativity. These research directions can provide valuable insights for educators and researchers, helping them develop effective teaching strategies.

9 | Implications

From a theoretical perspective, this study contributes to the growing body of literature on human co-creation with different partners in solving MCT, providing a foundation for discussing its application in other educational settings. The findings challenge the existing assumption that ChatGPT always surpasses human collaboration in terms of creativity, particularly in complex tasks like mathematical problem-solving, making a case for a more nuanced view of ChatGPT's role in creative work. This study also adds to the discourse on the cognitive mechanisms underlying creativity and emphasizes the importance of collaborative processes that involve human input. The finding that human–human co-creation is more accurate in assessing creativity further calls attention to the role of metacognition in creative processes, a key area for future research.

From a practical perspective, educators should emphasize using ChatGPT as a supportive tool, rather than a substitute for human collaboration. Teachers should encourage peer interaction among students to foster a more engaging and effective creative process. Furthermore, educators should introduce self-reflection activities to help students assess their creativity and problem-solving abilities more accurately. Students should not solely rely on internet information to support their learning, as it may not be sufficient to enhance their creativity. Therefore, teachers should create a co-creation learning environment that values both ChatGPT assistance and human collaboration. Educators should also adopt a balanced teaching strategy, exploring how to integrate ChatGPT into human–human collaboration while optimizing both forms of co-creation to generate more creative outcomes.

The present study found that the human–ChatGPT group enhanced the diversity and quantity of generated solutions, while the human–human group produced the most original solutions in solving MCT. These findings support the view that creativity is a unique human trait that is not easily replicated by ChatGPT. Furthermore, students working with peers accurately assessed their mathematical creativity, while those using ChatGPT overestimated it, and those using Google underestimated it. Also, students who worked with ChatGPT or a human peer found task resolution more interesting, easier, and less mentally demanding. They also found their co-creation partners more helpful and experienced greater self-efficacy. These findings will allow us to emphasize and enhance the potential of human–human co-creation, rather than reducing it through overreliance on ChatGPT. Future work should explore the specific social and cognitive mechanisms that drive the effectiveness of human–ChatGPT or human–human co-creation in MCT.

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Ethics Statement

The study was reviewed and approved by the Institutional Review Board affiliated with Yangzhou University. Participants over 18 years of age provided written informed consent directly, while for those under 18, additional written consent was obtained from their legal guardians.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.
Appendix. Supporting document.